

Bridging Grand Theory and AI Practice in Public Health: Through Middle-Range Theory, Demonstrated in PAPO, EquiRisk, and PMCO-AI

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Abstract

Artificial intelligence increasingly mediates public health interventions, but grand theory offers little guidance for building it — it explains population-level dynamics, not what a specific algorithm should do for one patient in one moment. This paper argues that middle-range theory is what bridges that gap: not by restating grand theory in different words, but by operationalizing the specific question it leaves unanswered into a construct precise enough to implement. We demonstrate this through three previously published frameworks — PAPO (policy-constrained action boundaries), EquiRisk (equity-aware risk stratification), and PMCO-AI (personalized motivation, capability, and opportunity) — each operationalizing a distinct concern at a different level, from macro policy to individual engagement. EquiRisk and PAPO have also been composed into one documented system, showing the translations can be layered; validating each framework independently, not the composition, is this paper's priority for future work.

Keywords: *middle-range theory; biomedical informatics; public health AI; policy-aware AI; algorithmic equity; behavioral informatics; theory-driven system design*

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1. Introduction

Public health has long relied on grand theoretical paradigms — the Social-Ecological Model (McLeroy et al., 1988), Self-Determination Theory (Deci & Ryan, 1985), structural accounts of health equity (World Health Organization, Commission on Social Determinants of Health, 2008) — to explain why interventions succeed or fail. These paradigms were built for human-mediated delivery: counseling, media campaigns, legislative action. They are, by design, high-abstraction and non-computational. Consider the difference concretely: the Social-Ecological Model's claim that policy and environment shape health outcomes is a general, qualitative statement, true across many settings but silent on any particular case. A computational requirement is different in kind — it must specify, for one AI agent about to issue one recommendation to one person, an answer to one question: is this specific action permitted for this specific patient, yes or no. Grand theory supplies claims of the first kind; software needs claims of the second, and grand theory alone does not produce them.

Robert Merton's concept of middle-range theory (Merton, 1968) names the space between minor hypotheses and grand theory: constructs specific enough to test and apply, general enough to transfer across settings. This paper treats bridging as a specific, repeatable move within that space, not a metaphor: state the grand theory's descriptive claim, name the operational question it leaves unanswered, then specify a middle-range construct precise enough to answer that question — and precise enough to be wrong. This is closer to operationalization in the measurement-theory sense than to translation as paraphrase.

We demonstrate the move through three previously published middle-range frameworks: PAPO (Policy-Aware Personalized Opportunity), EquiRisk (equity-aware risk stratification), and PMCO-AI (Personalized Motivation, Capability, and Opportunity in the AI era). Each operationalizes a different grand-theoretical concern, at a different level: PAPO at the policy/environment level, EquiRisk at the level of structural equity, PMCO-AI at the level of individual motivation and behavior.

2. Three Middle-Range Frameworks

All three frameworks originate in the author's Springer text on AI in public health (Wu, 2026e); each is illustrated below through the domain-specific SSRN paper that first applied it.

PAPO — Policy-Aware Personalized Opportunity. Illustrated for heatwave public health interventions (Wu, 2026b), PAPO addresses what its own paper terms institutional lag: public health policy and emergency protocols are developed through deliberative processes that move far slower than the AI-enabled environmental and health-monitoring systems now available in real time, leaving practitioners working from fragmented alerts and citizens uncertain which actions are actually feasible. Existing policy states, in general terms, who institutions should protect during a heatwave and how; it does not specify how that policy becomes one feasible, real-time action for one specific person. PAPO answers with a four-part architecture — a policy input layer that encodes institutional rules, a personalized opportunity engine that matches those rules to an individual's vulnerability profile and real-time context, a delivery interface that operationalizes the result through human-supervised channels, and a feedback loop that refines policy from aggregated outcomes — shifting the framework's own stated emphasis from risk signaling to opportunity creation. The paper positions PAPO explicitly as an instance of Merton's middle-range theory (Wu, 2026b, citing Merton, 1968; Boudon, 1991), operating between abstract policy principles and narrow technical implementation — independent testimony for this paper's own central claim.

EquiRisk — Equity-Aware Risk Stratification. Illustrated for diabetes management in food deserts (Wu, 2026a), EquiRisk starts from an observation about existing public-health infrastructure rather than a single named grand theory: validated instruments already exist for clinical risk (the HHS-HCC risk-adjustment model; Kautter et al., 2014), neighborhood vulnerability (the CDC/ATSDR Social Vulnerability Index, 2026), and area deprivation (the Neighborhood Atlas Area Deprivation Index; Kind & Buckingham, 2018), but these measures remain fragmented and are not combined into an operational decision rule. EquiRisk answers by integrating clinical risk, contextual vulnerability, and action feasibility into a single prioritization logic — developed further below into a multiplicative formula — shifting equity from a retrospective reporting metric into a proactive decision rule that lets a public health team answer, concretely, who can be helped now, and how.

PMCO-AI — Personalized Motivation, Capability, and Opportunity. Illustrated through dementia care (Wu, 2026c), PMCO-AI extends the COM-B model (Michie et al., 2011) with AI-driven personalization across each of its three components. COM-B states that capability, opportunity, and motivation jointly enable behavior; it does not say which to act on first when they conflict, or how a system should keep monitoring context after an initial plan is made. PMCO-AI answers with an explicit priority ordering — opportunity first — and an execution-layer AI system that continuously monitors context and updates its recommended options.

3. How the Three Relate: A Layered Reading

Read together, the three frameworks can be described structurally as nested layers: PAPO sets the outer, policy-facing boundary within which action is permitted; EquiRisk operates inside that boundary to determine equitable targeting; PMCO-AI operates inside the resulting space to execute personalized engagement with an individual. This is a structural reading of how the three relate on paper, not a claim that all three have been operated together — Section 4 turns to the one case where that composition has actually been documented. Table 1 summarizes the layer each framework occupies.

Layer	Anchor / Motivating Gap	Concern Addressed
PAPO	Institutional lag between slow policy and real-time systems; framed by its own paper as an instance of Merton's middle-range theory (Wu, 2026b, citing Merton, 1968; Boudon, 1991)	Generic risk alerts that fail to translate into feasible, real-time action
EquiRisk	Existing but uncombined risk/vulnerability instruments (HHS-HCC; CDC/ATSDR SVI; Neighborhood Atlas ADI)	Clinical-risk-only prioritization that overlooks contextual barriers to care
PMCO-AI	COM-B model (Michie et al., 2011)	Static, non-adaptive engagement that ignores motivational state and opportunities

Table 1. The layer each framework occupies, what motivates it, and the concern it addresses.

4. Illustrative Cases

The EquiRisk–PAPO case below receives more space than the PMCO-AI case for a specific reason: it is the one place among the three frameworks where a full, publicly documented specification already exists, so it can illustrate both the translation move and what a translated framework's operational content actually looks like. Its length reflects available documentation, not a claim that composition is this paper's validation priority — that priority, argued in Section 6, is testing each framework independently. The composition is documented and structurally interesting; it is not, on its own, evidence that composition should be validated ahead of the frameworks it composes.

EquiRisk and PAPO: Diabetes Care Navigation in Food Deserts. This translation has been carried past the conceptual stage: the author's CMDDS for EquiRisk-FoodDeserts v1.3 (Wu, 2026d) documents EquiRisk and PAPO as two phases of one implementable specification. EquiRisk computes priority multiplicatively — $\text{Priority} = \text{clinical_risk} \times (1 + \text{CVI}) \times \text{feasibility_weight}$ — combining clinical risk, contextual vulnerability, and action feasibility; PAPO then determines what is feasible and how it is delivered, checking policy eligibility and boundary conditions before assigning an opportunity or issuing

an explicit, escalated NULL when none exists. Consider two patients with comparable clinical urgency: for one, EquiRisk flags high need and PAPO matches a feasible pathway — a community health worker visit paired with a food-pharmacy referral; for the other, EquiRisk again flags high need, but no delivery-capable opportunity has remaining capacity this cycle, so PAPO returns an explicit NULL that escalates to a human case manager rather than assigning a poor-fit option or dropping the case silently. EquiRisk's own introducing paper (Wu, 2026a) independently illustrates the same logic with a comparable two-patient contrast — a food-desert and a non-food-desert patient with similar clinical risk — which this CMDDS-based version develops into a fully specified decision rule.

PMCO-AI: Dementia Care. PMCO-AI's own published case illustrates the same move at the individual level (Wu, 2026c). A dementia-care plan can look well-structured on paper — medications scheduled, meals planned, predictive alerts configured — and still diverge sharply in execution when a patient refuses a meal, wanders, or disengages from a reminder; a technically correct predictive alert offers no guidance for what to do next. PMCO-AI's response is opportunity-first execution: the system detects a calm moment and suggests a short walk or a preferred snack rather than insisting on the scheduled meal, gives the caregiver simplified guidance for tasks like medication administration, prompts brief engagement aligned with the patient's mood, and continuously revises its suggestions as circumstances change. The caregiver selects and adapts from the options offered; the system's contribution is narrowing what is feasible in the moment.

Neither case is empirical validation — both illustrate documented or published specifications, and the CMDDS itself designates its simulations as exploratory rather than predictive. Nor do the two cases show all three frameworks operating together: EquiRisk and PAPO are combined in one documented system, while PMCO-AI is illustrated independently. Validating each framework on its own terms, not integrating them into one pipeline, is this paper's priority (Section 6).

5. Implications and Generalizability

For system designers, the layered reading offers a design pattern that separates concerns often entangled in a single model: policy compliance, equitable targeting, and individual engagement. For evaluation science, it implies AI-mediated public health systems should be assessed against criteria at each layer independently, rather than a single aggregate metric that can mask failure at any one layer.

The translation move demonstrated here is not specific to these three grand theories. Public health draws on a much broader base — intrapersonal theories like the Health Belief Model (Rosenstock, 1974), community-level theories like Diffusion of Innovations (Rogers, 2003), and systems-level frameworks (Peters, 2014) — each of which has a grand-theoretical claim with no native computational representation. Some look like finer-grained decompositions of constructs PMCO-AI already models more coarsely. Others, like systems thinking, are arguably already translated: the EquiRisk–PAPO CMDDS's separation of tactical and strategic feedback loops is a direct computational instantiation of systems thinking's claim that acting on the wrong timescale produces instability, arrived at independently of this paper's argument. This is offered as a proposed research direction, not evidence that these translations have been built.

6. Limitations and Future Work

The priority for this research program is empirical validation of each middle-range framework independently — PAPO, EquiRisk, and PMCO-AI tested in their original domains — rather than validation of the three as a composed system. The EquiRisk-PAPO composition documented in the CMDDS (Wu, 2026d) is a structural observation this paper makes, not a research claim requiring its own validation.

- **PAPO** has not been empirically tested in a real heatwave deployment (Wu, 2026b); whether its policy-constrained action space prevents unsafe recommendations in a live system remains an open empirical question.
- **EquiRisk** has not been tested against real patient populations; the CMDDS's own simulations are exploratory rather than predictive.
- **PMCO-AI** has not been tested in a real deployment; its published case vignette (Wu, 2026c) is conceptual and qualitative, not a report of a live system.
- The EquiRisk-PAPO specification documents composability but does not validate either framework's individual claims; formal interoperability with PMCO-AI remains unspecified and is not a near-term priority.
- Generalizability beyond heatwave response, diabetes/food deserts, and dementia care has not been tested, and the broader translations proposed in Section 5 are unbuilt research directions.

7. Conclusion

Grand public health theory explains why intervention matters; algorithms determine how a specific data point is transformed. Between them sits a translation gap that AI-mediated public health systems repeatedly fall into during real-world deployment. This paper has argued, and illustrated through the EquiRisk-PAPO and PMCO-AI cases, that PAPO, EquiRisk, and PMCO-AI — though developed and validated independently — occupy exactly this middle-range space: each translates a specific piece of grand theory into computational logic. We offer this primarily as a demonstration of middle-range theory as a bridge between grand theory and AI practice. The priority for this research program is empirical validation of each framework individually; that the frameworks can be layered is a structural observation made along the way, not the research question being asked.

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Ethics Approval: This is a conceptual/perspective paper that illustrates published and documented frameworks using illustrative, non-identifiable case material; it does not involve human subjects research or identifiable patient data.

Data Availability: No new data was generated or analyzed in this conceptual paper.